

Milestone 3: Deterministic Baseline Simulation

Group 13

Scribe Question 1: System Objective and Deterministic Baseline

Project Objective

The primary objective of this project is to analyze the performance of a deterministic Binary Exponential Backoff (BEB) based medium access control mechanism in a multi-node wireless network environment using simulation.

In shared communication networks, multiple nodes attempt to transmit packets over the same communication channel. When more than one node transmits simultaneously, a collision occurs, resulting in packet loss and retransmission delays. Efficient channel access strategies are therefore required to regulate how nodes attempt to transmit packets.

The goal of the simulation is to study how the throughput and collision rate of the network change as the packet generation probability varies. Throughput represents the rate of successful transmissions, while collision rate represents how often simultaneous transmissions occur.

The simulation models a system consisting of:

- $N = 10$ nodes
- Each node generates packets probabilistically
- A shared communication medium
- A deterministic backoff-based collision resolution mechanism

The simulation runs for $T = 50,000$ time slots, allowing the system behavior to stabilize and meaningful statistical performance metrics to be observed.

The key performance metrics analyzed are:

1. Throughput

Number of successful transmissions per time slot.

2. Collision Rate

Number of collision events per time slot.

These metrics are evaluated as a function of the packet generation probability p , which varies from 0.01 to 0.5.

Deterministic Baseline Strategy

The baseline strategy implemented in this simulation is a deterministic version of the Binary Exponential Backoff (BEB) algorithm.

Binary Exponential Backoff is commonly used in network protocols such as Ethernet (CSMA/CD) and wireless MAC protocols. It is designed to reduce repeated collisions when multiple nodes compete for the same channel.

In this deterministic baseline:

1. Each node maintains a contention window (CW).
2. Each node also maintains a backoff counter.
3. A node can only transmit when its backoff counter reaches zero.

The operation proceeds as follows:

Packet Arrival

Packets arrive at nodes based on a deterministic schedule derived from the packet generation probability.

Backoff Countdown

If a node has packets waiting, it decreases its backoff counter each time slot.

Transmission Attempt

When the backoff counter becomes zero, the node attempts to transmit.

Transmission Outcomes

There are two possible outcomes:

Successful Transmission

If exactly one node transmits, the transmission succeeds. The node then:

- Removes the packet from its queue
- Resets its contention window to CW_MIN

Collision

If multiple nodes transmit simultaneously, a collision occurs. All transmitting nodes then:

- Double their contention window size
- Set a new deterministic backoff counter

Backoff Update Rule

The contention window follows

$$CW = \min(2 \times CW, CW_{MAX})$$

where

$$CW_{MIN} = 4$$

$$CW_{MAX} = 1024$$

The deterministic backoff value is computed using

$$backoff = (CW/2 + node_id) \bmod CW$$

This ensures nodes choose predictable backoff slots while avoiding randomness.

Why This Baseline is Reasonable

This deterministic baseline is reasonable for several reasons:

1. Binary Exponential Backoff is widely used. It is the foundation of many real-world MAC protocols.
2. Provides a controlled benchmark. Deterministic rules allow us to clearly observe system behavior without additional randomness.
3. Helps evaluate system dynamics. It shows how network performance changes as traffic load increases.
4. Prepares for randomized strategies. The deterministic model serves as a baseline to compare against more advanced stochastic strategies in later milestones.

Thus, this deterministic BEB implementation serves as a foundational model for studying medium access control under increasing network load.

Scribe Question 2: Random Variables in the Simulation

Although the decision mechanism in the simulation is deterministic, the system still contains random or probabilistic inputs that influence the behavior of the network.

These variables represent uncertainty in traffic generation and system load.

1 Packet Generation Process

Formal Definition

Let

$$p$$

represent the packet generation probability per time slot for each node.

This variable determines how frequently packets arrive at a node's transmission queue.

Distribution

Packet generation follows a Bernoulli process.

For each time slot:

$$P(\text{packet arrival}) = p$$

$$P(\text{no packet}) = 1 - p$$

This means that at every time slot a node independently generates a packet with probability p .

Implementation in Simulation

In the code, packet arrivals are approximated deterministically using

$$\text{arrival_interval} = \text{round}\left(\frac{1}{p}\right)$$

Packets are added to the queue whenever

$$(\text{slot} + \text{node_id}) \bmod \text{arrival_interval} = 0$$

This deterministic scheduling approximates the expected rate of a Bernoulli process.

2 Packet Queue Length

Formal Definition

Let

$$Q_i(t)$$

represent the number of packets waiting in the queue of node i at time t .

Distribution

The queue length evolves dynamically depending on

- Packet arrivals
- Successful transmissions
- Collisions

Thus the queue size follows a stochastic process influenced by traffic intensity and network contention.

Role in the Simulation

The queue length determines whether a node can attempt transmission.

A node can only transmit when

$$Q_i(t) > 0$$

If the queue becomes empty, the node stops participating in channel contention until another packet arrives.

3 Number of Transmitting Nodes

Formal Definition

Let

$$X_t$$

represent the number of nodes attempting to transmit in time slot t .

Distribution

Possible outcomes:

$$X_t = 0$$

Idle slot.

$$X_t = 1$$

Successful transmission.

$$X_t > 1$$

Collision occurs.

Role in the Simulation

The simulation determines the outcome of each slot by counting transmitting nodes.

- If $X_t = 1 \rightarrow$ success
- If $X_t > 1 \rightarrow$ collision

Scribe Question 3: Probabilistic Modeling and Fixed Decision Rule

The simulation combines probabilistic inputs with deterministic decision logic.

This approach allows the system to model real-world uncertainty while maintaining a predictable transmission policy.

Modeling Uncertainty

Uncertainty in the system arises from packet generation behavior.

Each node generates packets according to a probability parameter p .

Higher values of p represent higher traffic load, which increases network congestion.

Mathematical Representation

Packet generation is modeled using a Bernoulli process:

$$A_i(t) = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1 - p \end{cases}$$

where $A_i(t)$ indicates whether node i generates a packet at time t .

Queue Update Equation

The queue length evolves as

$$Q_i(t+1) = Q_i(t) + A_i(t) - S_i(t)$$

where

- $A_i(t)$ = packet arrival
- $S_i(t)$ = successful transmission

Deterministic Decision Rule

Each node follows a strict rule:

1. If queue is non-empty, decrement backoff counter.
2. If backoff counter reaches zero, attempt transmission.

The backoff value is determined using

$$backoff = (CW/2 + node_id) \bmod CW$$

Collision Handling

If multiple nodes transmit simultaneously

$$CW_i = \min(2 \times CW_i, CW_{MAX})$$

Even though the inputs contain uncertainty, the decision mechanism does not use random numbers.

Scribe Question 4: Simulation Pipeline

The simulation pipeline represents the complete sequence of operations executed during each time slot.

Step 1: Initialization

At the start of the simulation:

- Number of nodes $N = 10$
- Simulation duration $T = 50,000$
- Minimum contention window $CW_{MIN} = 4$
- Maximum contention window $CW_{MAX} = 1024$

Each node initializes:

- Queue length = 0
- Contention window = CW_{MIN}
- Deterministic backoff counter

Step 2: Input Generation

Packet arrival condition:

$$(slot + node_id) \bmod arrival_interval = 0$$

Step 3: Backoff Countdown

Nodes with pending packets decrement their backoff counters each slot.

Step 4: Transmission Attempt Detection

Nodes whose backoff becomes zero attempt transmission.

Step 5: Channel Outcome

Three cases:

- Idle Slot
- Successful Transmission
- Collision

Step 6: State Update

Nodes update contention windows after collisions.

Step 7: Performance Metrics

$$Throughput = \frac{successes}{T}$$

$$Collision\ Rate = \frac{collisions}{T}$$

Scribe Question 5: Simulation Outputs and Interpretation

The simulation generates performance metrics describing the efficiency of the deterministic BEB strategy.

Performance Metrics

1 Throughput

$$\text{Throughput} = \frac{\text{Successful Transmissions}}{\text{Total Slots}}$$

2 Collision Rate

$$\text{Collision Rate} = \frac{\text{Collision Events}}{\text{Total Slots}}$$

The following figures show the results obtained from the simulation.

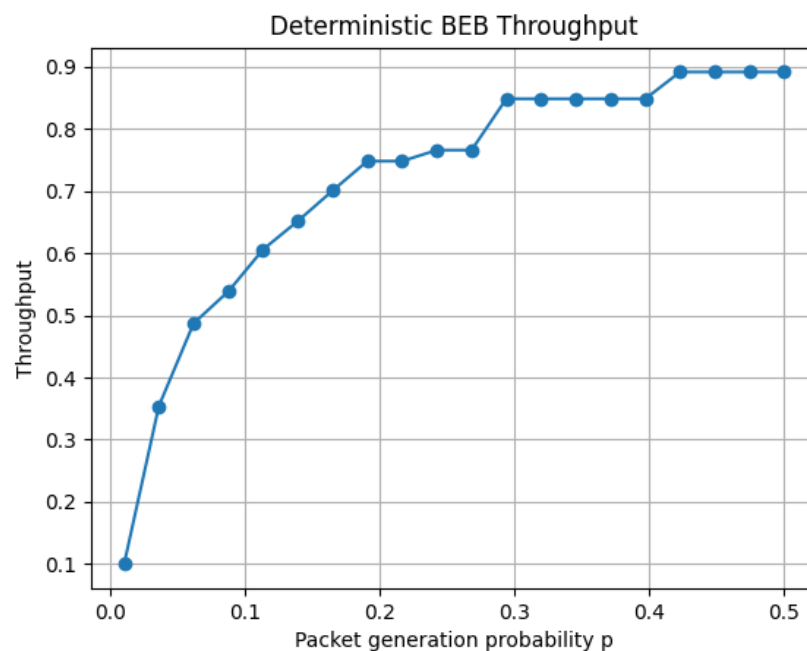


Figure 1: Throughput vs Packet Generation Probability

From the throughput graph we observe that when packet generation probability increases, the throughput initially increases because more packets are available for transmission.

However, after a certain point the number of collisions increases significantly and the throughput starts decreasing.

The collision graph shows that collision rate increases as traffic load increases because more nodes attempt transmission simultaneously.

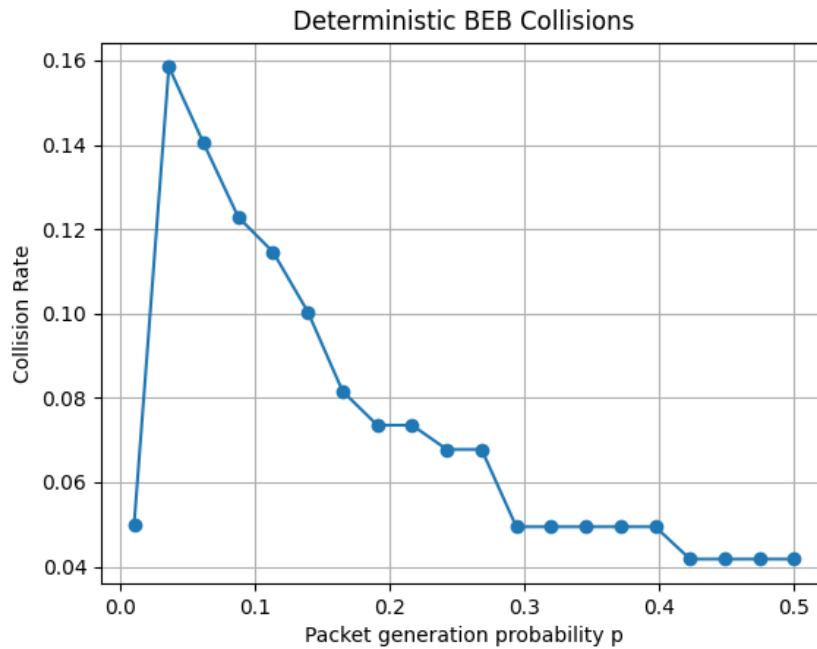


Figure 2: Collision Rate vs Packet Generation Probability

Observed pattern:

- Low traffic $\rightarrow$ rare collisions
- Medium traffic $\rightarrow$ collisions begin appearing
- High traffic $\rightarrow$ collision rate increases rapidly

Interpretation

There exists an optimal traffic load where throughput is maximized.

Beyond this point the network becomes congested and performance deteriorates.

Scribe Question 6: Technical Gaps and Preparation for Next Stage

While the deterministic baseline provides useful insights, the model still contains several simplifications.

Simplified Aspects

Deterministic Backoff Selection

$$backoff = (CW/2 + node_id) \bmod CW$$

In real protocols backoff values are random.

Simplified Traffic Model

Packet arrivals are approximated deterministically.

Ideal Channel Assumptions

The model assumes

- No transmission errors
- Perfect collision detection
- Instant feedback

Where Randomness Could Improve Performance

Possible improvements:

- Random backoff selection
- Random packet arrivals
- Randomized contention resolution

For example

$$backoff \sim Uniform(0, CW - 1)$$

Planned Improvements for Milestone 4

Future work may include

- Random backoff algorithms
- More realistic traffic models
- Adaptive transmission policies
- Comparison between deterministic and randomized strategies

These enhancements will allow the system to better represent real network behavior.